

GM-DAS-50/100 Distributed Acoustic / Vibration Interrogator

Flagship distributed vibration / acoustic platform · Φ -OTDR phase demodulation

Ultra-long range: 50 km standard, up to 100 km with optional Raman amplifier; wide band and massive channel count: 0.01 Hz~50 kHz response with 23,552 parallel channels; high fidelity: narrow-linewidth laser, waveform restoration $\geq 99\%$, quantitatively reconstructing the acoustic model for event classification. Unlike qualitative-only DVS, GM-DAS tells you what it is and where it is.



GM-DAS-50/100 interrogator

Oil & gas pipelines Perimeter security Rail protection zones
Seismic & geological survey Seepage imaging

Features

- 50 km standard, 100 km with optional Raman amplification
- 3.2 m spatial resolution, 23,552 parallel channels
- 0.01 Hz~50 kHz wide-band response (@1 km)
- Waveform restoration $\geq 99\%$ for intelligent event classification
- Narrow-linewidth laser: ≤ 1 kHz linewidth, RIN < -120 dB/Hz
- 250 MS/s · 14 bit acquisition, USB4.0 (Thunderbolt 4) throughput
- 2U rack chassis, drops straight into existing equipment rooms

Specifications

Optical & sensing performance

Measurement range	50 km standard / 100 km optional (Raman)
Spatial resolution	3.2 m
Frequency response	0.01 Hz~50 kHz @1 km
Demodulation channels	23,552 parallel
Waveform restoration	$\geq 99\%$
Laser linewidth / RIN	≤ 1 kHz · < -120 dB/Hz

Acquisition & interfaces

Data acquisition	250 MS/s · 14 bit
Optical channels	1~2
Data interface	USB4.0 (Thunderbolt 4)

Chassis & environment

Chassis	2U rack mount
Power supply	220 V AC

Operating temperature -20~60 °C

Ordering information

Model	Configuration	Notes
GM-DAS-50	50 km	Standard unit
GM-DAS-100	100 km	With Raman amplifier for ultra-long trunk lines
Custom	Channels / algorithm packs	Event-classification algorithms configured per industry

Field record

South-North Water Transfer middle route PCCP wire-break monitoring (results fully matched reality) · Hefei Metro Line 4, 44.8 km protection-zone alerting (5-minute closed loop) · G5011 expressway, 22 km (alerts 6 minutes ahead of emergency calls, accuracy 95%+).

Warranty & service

2 yr Hardware warranty · spares stocked	24 h On-site response, East China	48 h On-site response, nationwide	7×24 Remote diagnostics hotline
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Standard service policy; the supply contract prevails.

Certifications & standards

 CNAS testing	 CMA metrology	 ISO 9001 quality	 ISO 14001 environmental	 MA mining safety
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Certificate numbers and scans are available on request.